1. What is an Activity Diagram?

Activity Diagram

- An Activity Diagram is a UML behavioural diagram that models the DYNAMIC ASPECTS of a system by showing the flow of control or data from one activity (action) to another.
- It represents the sequence of activities performed in a use case, a business process, or any workflow.
- It is particularly useful for modelling workflows with DECISIONS, PARALLEL PATHS, and SWIM LANES showing who does what.

Key facts about Activity Diagrams:

- They show WHAT happens and in WHAT ORDER, unlike class diagrams which show structure.
- They are the UML equivalent of a flowchart, but more powerful.
- They can model concurrent (parallel) activities using Fork and Join nodes.
- Swim Lanes (also called partitions) show which ACTOR or COMPONENT is responsible for each activity.
- They are used during Requirements and Design phases to model use case flows.

2. Activity Diagram Notations

- An Activity Diagram uses the following standard UML notations. Every diagram is built using these symbols:



Solid filled black circle

Initial Node (Start)

Marks the STARTING POINT of the activity diagram. There is exactly ONE initial node per diagram. The flow begins here.



Circle with a filled dot inside

Activity Final Node (End)

Marks the END of the entire activity flow. When control reaches this node, the entire activity terminates. A diagram can have multiple final nodes.



Rounded rectangle

Activity (Action)

Represents a single STEP or ACTION in the process. Written as an action verb phrase (e.g., "Enter Login Credentials", "Submit Exam", "Generate Result").



Diamond shape

Decision Node

Represents a BRANCH POINT where the flow splits based on a condition. Has ONE incoming flow and TWO or MORE outgoing flows. Each outgoing arrow is labelled with a GUARD CONDITION in square brackets [ ].



Diamond shape (same as Decision)

Merge Node

Merges two or more alternate flows back into ONE. Has multiple incoming flows and ONE outgoing flow. Used to rejoin branches created by a Decision Node.



Thick horizontal/vertical bar

Fork Node

Splits ONE flow into MULTIPLE CONCURRENT (parallel) flows. Has one incoming and two or more outgoing arrows. Shows activities that can happen simultaneously.



Thick horizontal/vertical bar (same as Fork)

Join Node

SYNCHRONISES multiple parallel flows back into ONE. Has multiple incoming flows and one outgoing flow. The flow continues only after ALL incoming parallel flows complete.

Solid arrowed line

Control Flow (Arrow)

Shows the DIRECTION OF FLOW between activities, decisions, forks, and nodes. Can be optionally labelled with a condition or data name.



Vertical/Horizontal column with actor name at top

Swim Lane (Partition)

Divides the diagram into columns, each representing an ACTOR, ROLE, or SYSTEM COMPONENT. Activities in each lane belong to that actor. Swim lanes make it easy to see who does what.



Rectangle (without rounded corners)

Object Node

Represents an OBJECT or DATA that flows between activities. Used when you want to show that an activity produces or consumes a specific data object (optional, used in advanced diagrams).

3. Activity Diagram: Online Examination & Result Processing System

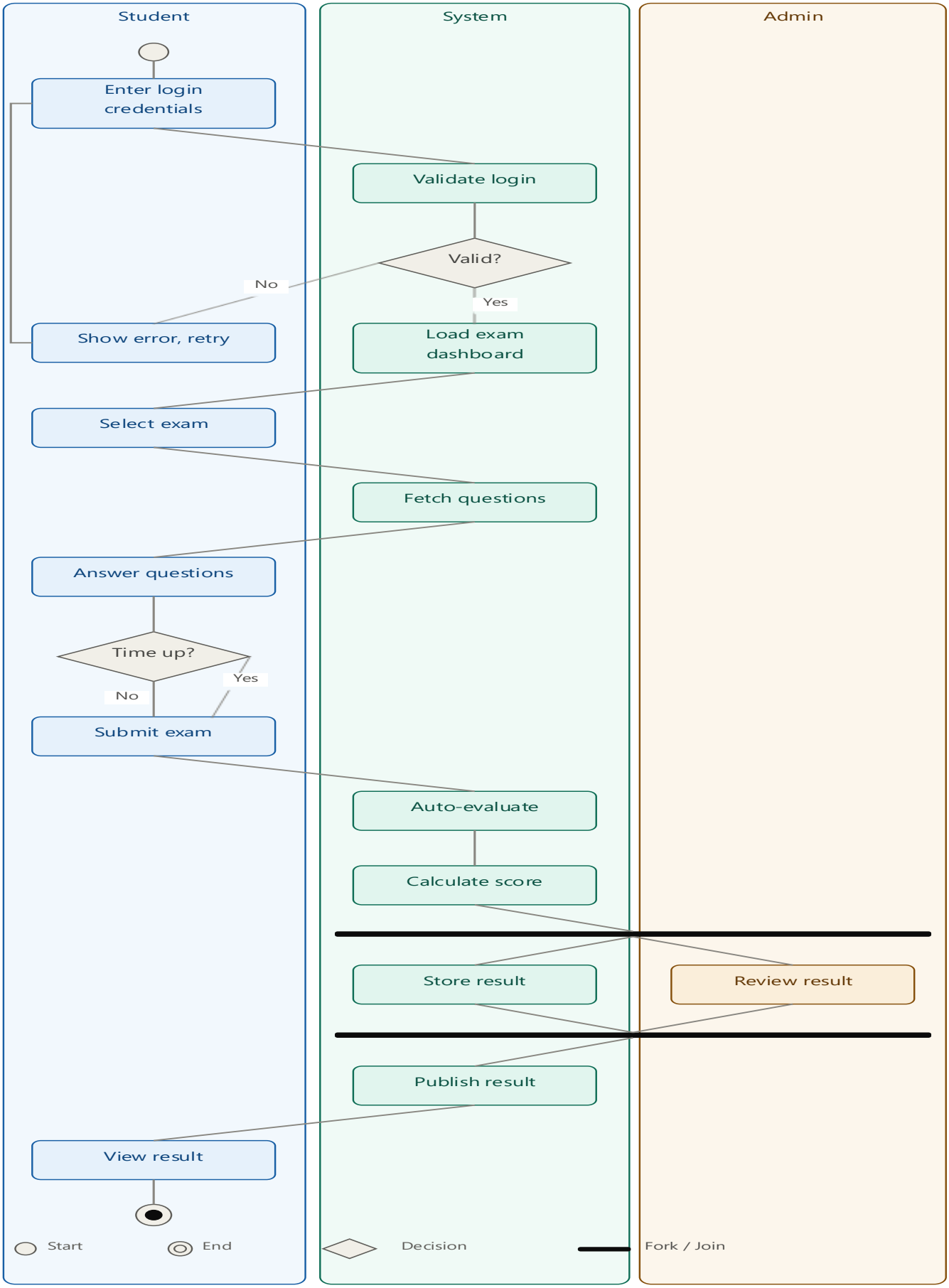
CASE STUDY 1: Online Examination & Result Processing System

System Overview

The Online Examination System allows students to log in, select and attempt exams, submit answers, and receive results automatically. Admins manage exam schedules and publish results. The key workflow to model is: the complete Student Exam Attempt Flow from login to result viewing.

Workflow 1: Student Login and Exam Attempt (with Swim Lanes)

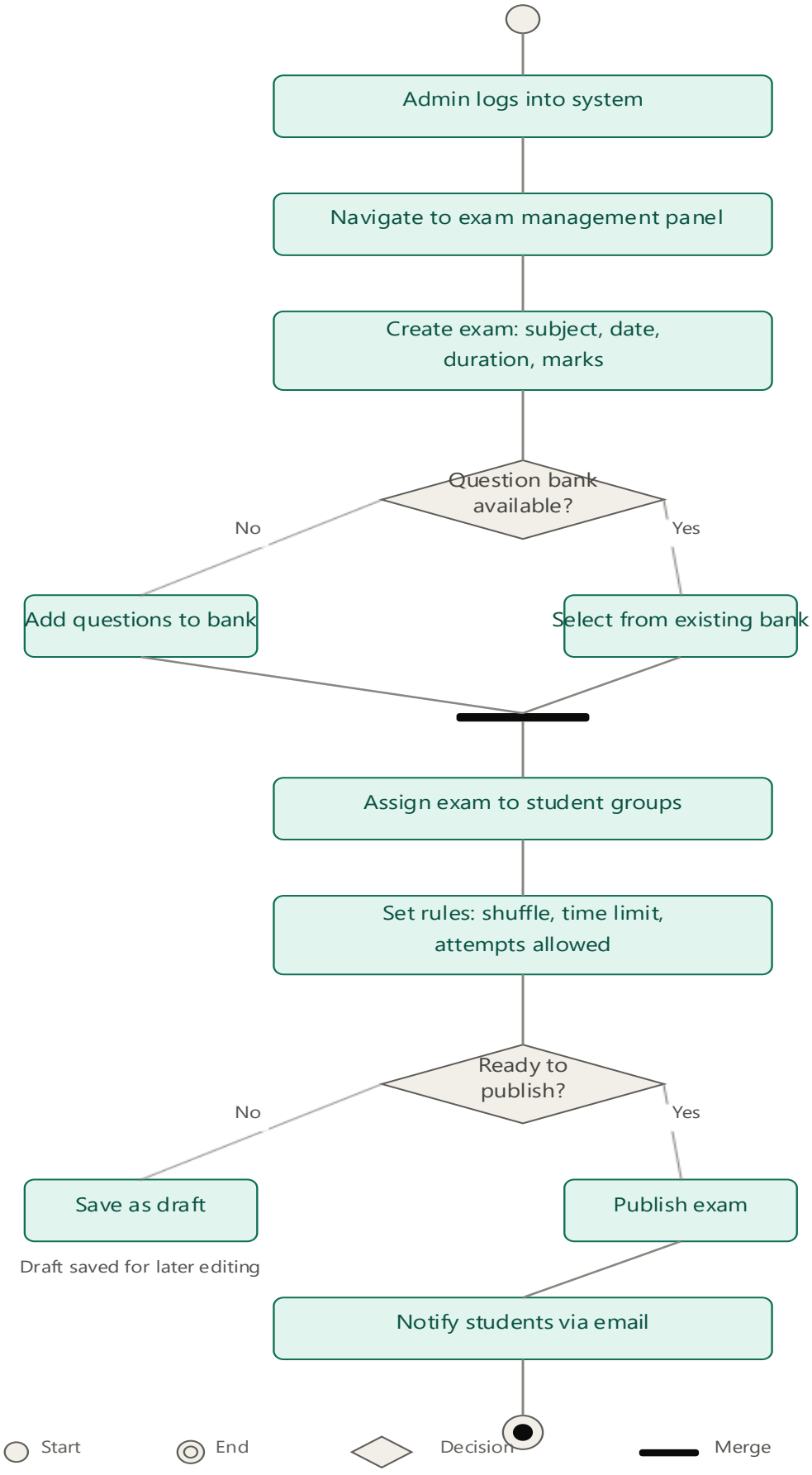
Actors / Swim Lanes: Student | System | Admin


Student
System
Admin
Enter login credentials
Validate login
Valid?
No
Yes
Show error, retry
Load exam dashboard
Select exam
Fetch questions
Answer questions
Time up?
Yes
No
Submit exam
Auto-evaluate
Calculate score
Store result
Review result
Publish result
View result
Start
End
Decision
Fork / Join

Step-by-Step Explanation of the Diagram

Step	Activity & Notation Used
1. Start	Initial Node (●) — placed in Student swim lane. The process begins when the student opens the exam portal.
2. Enter Login	Activity node: "Enter Login Credentials" — student types username and password.
3. Validate Login	Activity node in SYSTEM lane: "Validate Login" — system checks credentials against the Student DB.
4. Decision	Decision Node (◇): [Valid?] — If NO → Show Error / Retry. If YES → Load Exam Dashboard.
5. Select Exam	Activity: "Select Exam" — student picks the exam from the list.
6. Fetch Paper	Activity in SYSTEM lane: "Fetch Question Paper" — system loads questions from Question Bank DB.
7. Attempt Exam	Activity: "Answer Questions" — student answers one by one within the time limit.
8. Decision	Decision Node (◇): [Time Up?] — If YES → Auto-Submit. If NO → Manual Submit.
9. Auto Evaluate	Activity in SYSTEM lane: "Auto Evaluate" — system compares answers with Answer Key.
10. Publish	Fork: System stores result AND Admin reviews result (parallel). Join: Both must complete before result is published.
11. View Result	Activity in STUDENT lane: "View Result" — student sees score, grade, pass/fail.
12. End	Final Node (◉) in Student lane — workflow ends.

Workflow 2: Admin Creates and Schedules Exam


Admin logs into system
Navigate to exam management panel
Create exam: subject, date, duration, marks
Question bank available?
No
Yes
Add questions to bank
Select from existing bank
Assign exam to student groups
Set rules: shuffle, time limit, attempts allowed
Ready to publish?
No
Yes
Save as draft
Draft saved for later editing
Publish exam
Notify students via email
Start
End
Decision
Merge